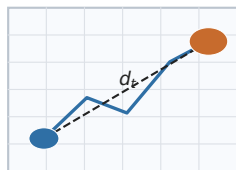


● robot / microrobot ● patient hand

a

Simulation environment



CMD-DR virtual patient

intent: SPC / learned hand

motor layer: delay, noise, LPF, accel. limit

ZPD reward + boundary

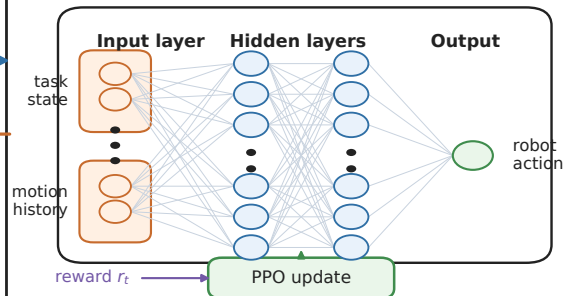
O_t, r_t

a_t

same task

b

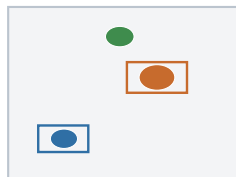
Adaptive policy



trained policy

c

Vision and state estimation



async vision thread

YOLO microrobot

MediaPipe hand

state buffer +
dead reckoning

image feed

real O_t

d

Real-world rehabilitation platform

